Paper 16 - Continuum Edge Residuals

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Construction Basis

Use powers of ten and edge residuals as practical windows into unsolved continuum depth.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-16.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-16\paper_kernel_manifest.json
- body: production\papers\CQE-paper-16\PAPER-BODY.md
- source: production\papers\CQE-paper-16\SOURCE.md
- formal: production\papers\CQE-paper-16\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-16\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-16\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-16.md

Paper 16 - Continuum Edge Residuals

Abstract

This paper defines continuum edge residuals as local window receipts. The closed result is local: every chart state anneals to a Lie-conjugate rest state in at most three `S3` steps, the edge residue is exactly the state condition `C=1, R=0`, and power-of-ten windows provide a practical way to sample the boundary between resolved interior and newly exposed depth. The global continuum limit is not closed here. The collapse of the propagating correction sum from `O(N)` to `O(log N)` remains the McKay-Thompson parity obligation.

Claims

Claim 16.1. Every local chart state closes to a Lie-conjugate rest state in at most three `S3` transposition steps.

Claim 16.2. There are exactly four Lie-conjugate rest states.

Claim 16.3. Edge residue is exactly `C AND NOT R`, so it fires only at `(0,1,0)` and `(1,1,0)`.

Claim 16.4. Power-of-ten windows are valid local receipt windows.

Claim 16.5. Local/oracle nth-bit checks pass with correction included, but the global correction collapse remains open.

Definitions

A **rollout** is the local process of reading a state until it reaches rest.

A **Lie-conjugate rest state** is an `L=R` chart state.

An **edge residual** is a carry in flight at a window boundary:

```
edge_residue(L,C,R) = C AND NOT R
```

A **power-of-ten window** is a practical aperture at depths `10`, `100`, `1000`, and so on. It is a receipt window, not a continuum proof.

Theorem 16

Continuum edge residuals are locally well-defined window receipts:

```
local state -> <=3-step rest closure -> edge_residue = C AND NOT R
```

and every global continuum claim remains an obligation until the propagating correction sum is closed.

Proof

The centroid verifier checks all eight chart states and reports local closure. Every state anneals to a Lie-conjugate rest state in at most three `S3` steps. This proves Claim 16.1.

The rest states are the four states satisfying `L=R`. The verifier reports the count as `4`. This proves Claim 16.2.

The edge-residue formula is `C AND NOT R`. Exhausting all eight states gives exactly `(0,1,0)` and `(1,1,0)`. This proves Claim 16.3.

The verifier samples windows at `10`, `100`, and `1000`. For each window it records the selected local state, edge-residue value, anneal step count, and final rest state. Each sampled window closes locally. This proves Claim 16.4 as a local receipt statement.

The nth-bit layer passes with local/oracle correction included, but the receipt names McKay-Thompson correction parity as open. Therefore the local edge residual is admitted while global continuum collapse is not. This proves Claim 16.5.

Together these claims prove the theorem.

Receipt

Promoted verifier:

```
production/formal-papers/CQE-paper-16/verify_continuum_edge_residuals.py
```

Receipt:

```
production/formal-papers/CQE-paper-16/continuum_edge_residuals_receipt.json
```

Closed layers:

```
every local chart state anneals to a Lie-conjugate rest state in <=3 S3 steps
there are four Lie-conjugate rest states
edge residue is exactly C and not R
sample decade windows carry local receipts
local/oracle nth-bit layer passes with correction included
```

Open layers:

```
global continuum closure
O(N) to O(log N) propagating-correction collapse
closed McKay-Thompson correction parity
claim that adding digits terminates continuum depth
```

Falsifiers

The paper fails if any local chart state needs more than three anneal steps.

It fails if edge residue fires outside `C=1, R=0`.

It fails if power-of-ten windows are treated as a completed continuum limit.

It fails if the McKay-Thompson parity obligation is hidden.

Role in the Suite

Paper 15 exports a residue carrier. Paper 16 turns that carrier into a windowed edge reading. Paper 17 may use the windowed residual as an input to the error-correction tower only by carrying the open global-collapse boundary.

Conclusion

Paper 16 closes the local edge-residual instrument. It does not close the continuum limit. That distinction lets later papers use powers of ten as inspectable windows while preserving the deeper obligation.